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Dominating K_t -ModelsFreddie Illingworth¹  | David R. Wood² ¹Department of Mathematics, University College London, London, UK | ²School of Mathematics, Monash University, Melbourne, Australia**Correspondence:** David R. Wood (david.wood@monash.edu)**Received:** 22 July 2024 | **Revised:** 21 March 2025 | **Accepted:** 30 May 2025**Funding:** Research supported by EPSRC grant EP/V521917/1 and the Heilbronn Institute for Mathematical Research. Research supported by the Australian Research Council.**Keywords:** dominating set | graph minor | graph theory | Hadwiger's conjecture | random graph

ABSTRACT

A *dominating K_t -model* in a graph G is a sequence $(T_1, \dots, T_t)$ of pairwise disjoint non-empty connected subgraphs of G , such that for $1 \leq i < j \leq t$ every vertex in T_j has a neighbour in T_i . Replacing ‘every vertex in T_j ’ by ‘some vertex in T_j ’ retrieves the standard definition of K_t -model, which is equivalent to K_t being a minor of G . We explore in what sense dominating K_t -models behave like (non-dominating) K_t -models. The two notions are equivalent for $t \leq 3$ but are already very different for $t = 4$, since the 1-subdivision of any graph has no dominating K_4 -model. Nevertheless, we show that every graph with no dominating K_4 -model is 2-degenerate and 3-colourable. More generally, we prove that every graph with no dominating K_t -model is 2^{t-2} -colourable. Motivated by the connection to chromatic number, we study the maximum average degree of graphs with no dominating K_t -model. We give an upper bound of 2^{t-2} and show that random graphs provide a lower bound of $(1 - o(1))t \log t$, which we conjecture is asymptotically tight. This result is in contrast to the K_t -minor-free setting, where the maximum average degree is $\Theta(t\sqrt{\log t})$. A natural strengthening of Hadwiger's conjecture arises: Is every graph with no dominating K_t -model $(t - 1)$ -colourable? We provide two pieces of evidence for this: (1) It is true for almost every graph. (2) Every graph G with no dominating K_t -model has a $(t - 1)$ -colourable induced subgraph on at least half the vertices, which implies there is an independent set of size at least $\frac{|V(G)|}{2t-2}$.

1 | Introduction

A *dominating K_t -model* in a graph G is a sequence $(T_1, \dots, T_t)$ of pairwise disjoint non-empty connected subgraphs of G , such that for $1 \leq i < j \leq t$ every vertex in T_j has a neighbour in T_i . Each T_i dominates $T_{i+1} \cup \dots \cup T_t$, hence the name.¹ Contracting each T_i into a single vertex and deleting vertices not in $T_1 \cup \dots \cup T_t$ gives K_t as a minor. Indeed, in the definition of dominating K_t -model, replacing ‘every vertex in T_j ’ by ‘some vertex in T_j ’ retrieves the standard definition of K_t -model, which is equivalent to K_t being a minor of G .

At first glance, the definition of dominating K_t -model might seem very restrictive. This paper explores in what sense dominating K_t -models behave like (non-dominating) K_t -models. We

show that several proof methods in the literature regarding K_t -models in fact work with dominating K_t -models. On the other hand, we show some significant differences.

For $t \leq 3$ it is easily seen that a graph G has a K_t -minor if and only if G has a dominating K_t -model (Observation 1). However, for $t = 4$, the behaviour changes dramatically: There are graphs that contain arbitrarily large complete graph minors, but contain no dominating K_4 -model (Corollary 6). Nevertheless, we show that every graph with minimum degree at least three contains a dominating K_4 -model (Theorem 7), thus strengthening a classical result of Hadwiger [1] and Dirac [2].

Now consider the chromatic number $\chi(G)$. Hadwiger [1] famously conjectured that every K_t -minor-free graph is

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$(t - 1)$ -colourable. This is widely considered to be one of the most important open problems in combinatorics (see [3] for a survey). The best upper bound on the chromatic number is $\mathcal{O}(t \log \log t)$ due to Delcourt and Postle [4]. It is open whether K_t -minor-free graphs are $\mathcal{O}(t)$ -colourable.

The following natural question arises: What is the maximum chromatic number of a graph with no dominating K_t -model? We prove that this maximum exists, and in particular, the answer is at most $3 \cdot 2^{t-4}$ for $t \geq 4$ (see Theorem 22). Complete graphs provide a lower bound of $t - 1$. It is possible that every graph with no dominating K_t -model is $(t - 1)$ -colourable, which would be a considerable strengthening of Hadwiger's conjecture. We prove that this is true for $t \leq 4$ (see Corollary 9).

We show that two pieces of evidence for Hadwiger's conjecture also hold for its dominating version. First, consider independent sets. Let $\alpha(G)$ be the maximum size of an independent set in a graph G . Hadwiger's conjecture would imply that $\alpha(G) \geq \frac{n}{t-1}$ for every n -vertex K_t -minor-free graph G . Duchet and Meyniel [5] proved that $\alpha(G) \geq \frac{n}{2t-2}$ for such G . We extend this result for graphs with no dominating K_t -model (Theorem 14).

The second piece of evidence concerns random graphs. The *Hadwiger number* of a graph G , denoted by $\text{had}(G)$, is the maximum integer t such that K_t is a minor of G . Hadwiger's conjecture asserts that $\chi(G) \leq \text{had}(G)$ for every graph G . Similarly, define the *dominating Hadwiger number* of a graph G , denoted by $\text{domhad}(G)$, to be the maximum integer t such that G contains a dominating K_t -model.

Bollobás et al. [6] showed that almost every graph G satisfies $\chi(G) \leq \text{had}(G)$. This means that the probability that a random n -vertex graph satisfies $\chi(G) \leq \text{had}(G)$ tends to 1 as $n \rightarrow \infty$. Strengthening the result of Bollobás et al. [6], we show that almost every graph G satisfies $\chi(G) \leq \text{domhad}(G)$ (Corollary 23). The proof of this result shows that the Hadwiger number and the dominating Hadwiger number behave differently for the Erdős-Rényi² random graph $G(n, p)$. In particular, $\text{domhad}\left(G\left(n, \frac{1}{2}\right)\right) = (1 + o(1))\frac{n}{\log_2 n}$, whereas $\text{had}\left(G\left(n, \frac{1}{2}\right)\right) = (1 + o(1))\frac{n}{\sqrt{\log_2 n}}$.

Random graphs and the chromatic number both relate to the natural extremal questions: What is the maximum average degree of a K_t -minor-free graph? Kostochka [7, 8] and Thomason [9] independently answered this question, by showing that the maximum average degree of a K_t -minor-free graph is in $\Theta(t\sqrt{\log t})$, where random graphs provide the extremal examples. Later, Thomason [10] determined the leading constant asymptotically. Consider the analogous question for dominating models: What is the maximum average degree of a graph with no dominating K_t -model? We show that the answer is at most 2^{t-2} (Theorem 16) and that random graphs provide a lower bound of $(1 - o(1))t \log t$. Closing this gap is an interesting open problem. We establish a number of results that suggest a $\mathcal{O}(t \log t)$ upper bound.

This paper is structured as follows: Section 2 presents some basic observations, comparing K_t -models and dominating K_t -models. Section 3 focuses on the structure of graphs with no

dominating K_4 -model. Section 4 studies dominating K_t -models in random graphs. Section 5 studies the structure of graphs with no dominating K_t -model and concludes bounds on their independence number. Sections 6 and 7 respectively study the average degree and chromatic number of graphs with no dominating K_t -model. Section 8 concludes with a number of open problems.

2 | Basics

First note that K_t -models and dominating K_t -models are equivalent for $t \leq 3$.

Observation 1. For $t \in \{1, 2, 3\}$, a graph G has a dominating K_t -model if and only if G has a K_t -model.

Proof. The $t = 1$ case holds since a graph has a dominating K_1 -model if and only if $V(G) \neq \emptyset$ if and only if G has a K_1 -model. The $t = 2$ case holds since a graph has a dominating K_2 -model if and only if $E(G) \neq \emptyset$ if and only if G has a K_2 -model. The $t = 3$ case holds since a graph has dominating K_3 -model if and only if G has a cycle if and only if G has a K_3 -model. In particular, if vw is an edge of a cycle C , then $(C - v - w, \{v\}, \{w\})$ is a dominating K_3 -model. $\square$

Note the following elementary observations.

Observation 2. For $t \geq 2$, a graph G has a dominating K_t -model if and only if G has a non-empty connected subgraph T such that $N_G(T)$ has a dominating K_{t-1} -model.

Observation 3. For $t \geq 2$, if a graph G has a dominating K_t -model, then G has a K_t -model $(T_1, \dots, T_t)$ where each of T_{t-1} and T_t has exactly one vertex.

Proof. This follows from Observation 2 and the fact that a graph has a dominating K_2 -model if and only if it contains an edge. $\square$

We remark that Martinsson and Steiner (personal communication, 2023) conjectured that every graph with chromatic number t contains a K_t -minor with two singleton branch sets. By Observation 3, their conjecture lies between Hadwiger's original conjecture and the dominating version.

Lemma 4. If a graph G has a dominating K_t -model, then G has a path $v_1, \dots, v_t$ where $\deg_G(v_1) \geq t - 1$, and $\deg_G(v_i) \geq i$ for $1 \leq i \leq t - 1$.

Proof. Say $(T_1, \dots, T_t)$ is a dominating K_t -model in G . Let v_t be any vertex in T_t . Then v_t has a neighbour in each of $T_1, \dots, T_{t-1}$, so $\deg_G(v_t) \geq t - 1$, as claimed. For $i = t - 1, t - 2, \dots, 1$, let v_i be a vertex in T_i adjacent to v_{i+1} . Each v_i has a neighbour in each of $T_1, \dots, T_{i-1}, T_{i+1}$, so $\deg_G(v_i) \geq i$. By construction, $v_1, \dots, v_t$ is a path. $\square$

Lemma 4 implies the following.

Corollary 5. Every graph with maximum degree Δ contains no dominating $K_{\Delta+2}$ -model.

Corollary 5 highlights a big difference between K_t -models and dominating K_t -models, since there are n -vertex graphs with maximum degree 3 that contain $K_{c\sqrt{n}}$ -models but contain no dominating K_5 -model by Corollary 5. This difference is also highlighted by the following result, which is an immediate consequence of Lemma 4.

Corollary 6. *For $n \geq 4$, if G is any graph obtained from K_n by subdividing each edge at least once, then G contains no dominating K_4 -model.*

We conclude this section by mentioning a curious consequence of Corollary 5. If $\chi(G) \leq \text{domhad}(G)$ for every graph G , then Corollary 5 would imply that every graph with maximum degree Δ is $(\Delta + 1)$ -colourable, which is true since such graphs are Δ -degenerate.

3 | Dominating K_4 -Models

Hadwiger [1] and Dirac [2] proved that every graph with minimum degree at least 3 has a K_4 -model. Here we strengthen this result.

Theorem 7. *Every graph G with minimum degree at least 3 has a dominating K_4 -model.*

Proof. We may assume that G is connected. Since $\delta(G) \geq 3$, G has a cycle. Choose a pair (C, H) where C is a cycle of G and H is a component of $G - C$ such that $|V(H)|$ is maximum (over all choices of (C, H)) and, subject to this, $|V(C)|$ is minimum. By minimality, C is an induced cycle of G . Since $\delta(G) \geq 3$, H is non-empty. Since G is connected, there is a vertex z of C with at least one neighbour in H .

Suppose that $G - (V(H) \cup \{z\})$ contains a cycle C' . Then $G - C'$ has a component containing $\{z\} \cup V(H)$, which contradicts the choice of (C, H) . Thus

$$G' := G - (V(H) \cup \{z\}) \text{ is a forest.} \quad (\dagger)$$

Let x and y be the neighbours of z in C and P be the path $C - \{x, y, z\}$. All the vertices of P have degree at least two in $C - z$ and so also in G' . All the vertices of $G - (C \cup H)$ have no neighbours in H and so have degree at least two in G' . Hence, in G' , every vertex has degree at least two, except possibly x and y . But G' is a forest and so, in G' , both x and y have degree one, and every other vertex of P has degree two. Therefore, since C is an induced cycle of G and $\delta(G) \geq 3$, every vertex of C has a neighbour in H . Then $(H, P + y, \{x\}, \{z\})$ is a dominating K_4 -model in G . $\square$

Theorem 7 implies the following result.

Corollary 8. *For $n \geq 2$, every n -vertex graph with no dominating K_4 -model has at most $2n - 3$ edges.*

Hadwiger [1] and Dirac [2] proved that every K_4 -minor-free graph is 3-colourable. Theorem 7 implies the following strengthening.

Corollary 9. *Every graph with no dominating K_4 -model is 2-degenerate and 3-colourable.*

These two results suggest that dominating K_4 -models behave like K_4 -models, although we emphasise that there are graphs with no dominating K_4 -model that contain arbitrarily large complete graph minors (Corollary 6).

4 | Random Graphs

The main result of this section (Theorem 11) asymptotically determines the dominating Hadwiger number of $G(n, p)$. We will ignore ceilings and floors in this section. We first need the following lemma.

Lemma 10. *For fixed $p \in (0, 1)$,*

$$\mathbb{P}(G(n, p) \text{ is not connected}) = \mathcal{O}(n(1 - p)^n),$$

where the implied constant may depend on p but not on n .

Proof. Say a set S of vertices in $G(n, p)$ is *isolated* if there are no edges between S and the rest of the vertices. For a fixed set S of k vertices,

$$\mathbb{P}(S \text{ is isolated}) = (1 - p)^{k(n-k)}.$$

If $G(n, p)$ is not connected, then there is some non-empty set of at most $n/2$ vertices that is isolated. Thus, taking a union bound and using the inequality $\binom{n}{k} \leq n^k$ gives

$$\begin{aligned} \mathbb{P}(G(n, p) \text{ is not connected}) &\leq \sum_{k=1}^{n/2} \binom{n}{k} (1 - p)^{k(n-k)} \\ &\leq n(1 - p)^{n-1} + n^2(1 - p)^{2(n-2)} \\ &\quad + \sum_{k=3}^{n/2} (n(1 - p)^{n-k})^k \\ &\leq \mathcal{O}(n(1 - p)^n) \\ &\quad + \sum_{k=3}^{n/2} (n(1 - p)^{n/2})^k \\ &\leq \mathcal{O}(n(1 - p)^n) + \frac{(n(1 - p)^{n/2})^3}{1 - n(1 - p)^{n/2}} \\ &= \mathcal{O}(n(1 - p)^n). \quad \square \end{aligned}$$

Theorem 11. *For constant $p \in (0, 1)$, asymptotically almost surely,*

$$\text{domhad}(G(n, p)) = (1 + o(1)) \frac{n}{\log_b n},$$

where $b = \frac{1}{1-p}$.

Proof. We first prove the lower bound. Fix $\varepsilon > 0$ and let $t = \frac{n}{(1+\varepsilon)\log_b n}$. Partition $V(G(n, p))$ into t parts, $V_1, \dots, V_t$, each of

size $n/t = (1 + \varepsilon) \log_b n$. We call a part V_i *good* if the subgraph of $G(n, p)$ induced by V_i is connected, and a pair (V_i, V_j) (with $i < j$) *good* if $V_j \subseteq N(V_i)$.

Note that $(1 - p)^{n/t} = n^{-1-\varepsilon}$. Since the graph induced by V_i is $G(n/t, p)$, Lemma 10 gives

$$\mathbb{P}(V_i \text{ is bad}) = \mathcal{O}((1 + \varepsilon)(\log_b n)n^{-1-\varepsilon}).$$

By the union bound, the probability that some V_i is bad is $\mathcal{O}((1 + \varepsilon)(\log_b n)n^{-\varepsilon}) = o(1)$. Hence, with high probability, there are no bad parts.

The probability that a fixed vertex in V_j has no neighbour in V_i is $(1 - p)^{|V_i|}$. Thus,

$$\mathbb{P}((V_i, V_j) \text{ is good}) = (1 - (1 - p)^{|V_i|})^{|V_j|}. \quad (1)$$

Using the inequality $(1 + x)^r \geq 1 + rx$ for $r \geq 1$ and $x \geq -1$ (this is Bernoulli's inequality) we obtain

$$\mathbb{P}((V_i, V_j) \text{ is bad}) = 1 - (1 - n^{-1-\varepsilon})^{n/t} \leq n^{-\varepsilon} t^{-1}.$$

Hence, the expected number of bad pairs of parts is at most

$$\binom{t}{2} \cdot n^{-\varepsilon} t^{-1} \leq t n^{-\varepsilon}.$$

By Markov's inequality, the probability that there are at least εt bad pairs is at most $\varepsilon^{-1} n^{-\varepsilon} = o(1)$. Thus, with high probability, the number of bad pairs is at most εt , and there are no bad parts. In this case, deleting one part from each bad pair leaves a dominating clique model. Thus, with high probability, $G(n, p)$ contains a dominating K_s -model where

$$s \geq (1 - \varepsilon)t = \frac{1 - \varepsilon}{1 + \varepsilon} \cdot \frac{n}{\log_b n} \geq (1 - 2\varepsilon) \frac{n}{\log_b n}.$$

We now prove the upper bound. Define a *dominating pseudo- K_t -model* in a graph G to be a sequence $(V_1, \dots, V_t)$ of pairwise disjoint non-empty subsets of $V(G)$, such that $V_j \subseteq N(V_i)$ for $1 \leq i < j \leq t$ (this is just a dominating K_t -model without the connectedness condition).

Fix $\varepsilon > 0$ and take $t - 1 = (1 + \varepsilon) \frac{n-1}{\log_b n}$. It suffices to show that $G(n, p)$ does not contain a dominating pseudo- K_t -model with high probability. If $(V_1, \dots, V_t)$ is a dominating pseudo- K_t -model and $v \in V_t$, then $(V(G) \setminus (V_2 \cup \dots \cup V_{t-1} \cup \{v\}), V_2, \dots, V_{t-1}, \{v\})$ is also a dominating pseudo- K_t -model. Thus, it suffices to show that, with high probability, $G(n, p)$ does not contain a dominating pseudo- K_t -model where $|V_t| = 1$ and $V_1 \cup \dots \cup V_t = V(G(n, p))$.

Fix a list of non-empty vertex sets $(V_1, \dots, V_t)$ that partition $V(G(n, p))$ and such that $|V_t| = 1$. Then $(V_1, \dots, V_t)$ is a dominating pseudo- K_t -model if and only if (V_i, V_j) is good for all $1 \leq i < j \leq t$. By Equation (1), and since $1 - x \leq e^{-x}$ for $x \geq 0$,

$\mathbb{P}((V_1, \dots, V_t)$ is a dominating pseudo- K_t -model)

$$\begin{aligned} &= \prod_{1 \leq i < j \leq t} (1 - (1 - p)^{|V_i|})^{|V_j|} \\ &\leq \prod_{1 \leq i < j \leq t} \exp\{-|V_j|(1 - p)^{|V_i|}\} \\ &= \exp\left\{-\sum_{1 \leq i < j \leq t} |V_j|(1 - p)^{|V_i|}\right\}. \end{aligned}$$

We now prove a lower bound for the sum in the parentheses. Since each V_j has size at least 1,

$$\sum_{1 \leq i < j \leq t} |V_j|(1 - p)^{|V_i|} \geq \sum_{1 \leq i < j \leq t} (1 - p)^{|V_i|} = \sum_{i=1}^{t-1} (t - i) \cdot (1 - p)^{|V_i|}.$$

The Arithmetic Mean-Geometric Mean inequality and the fact that $|V_i| = 1$ gives

$$\begin{aligned} \sum_{i=1}^{t-1} (t - i)(1 - p)^{|V_i|} &\geq (t - 1) \left[\prod_{i=1}^{t-1} (t - i)(1 - p)^{|V_i|} \right]^{1/(t-1)} \\ &= (t - 1) \left[(t - 1)! (1 - p)^{\sum_{i=1}^{t-1} |V_i|} \right]^{1/(t-1)} \\ &= (t - 1) \left[(t - 1)! \right]^{\frac{1}{t-1}} (1 - p)^{(n-1)/(t-1)} \\ &\geq \frac{(t - 1)^2}{e} (1 - p)^{(n-1)/(t-1)}, \end{aligned}$$

where the final inequality used the lower bound $k! > \left(\frac{k}{e}\right)^k$, which is valid for all positive integers k . In particular,

$\mathbb{P}((V_1, \dots, V_t)$ is a dominating pseudo- K_t -model)

$$\leq \exp\left\{-\frac{(t - 1)^2}{e} (1 - p)^{(n-1)/(t-1)}\right\}.$$

Since the number of partitions of $V(G(n, p))$ into t parts is t^n ,

$\mathbb{E}(\# \text{ of dominating pseudo-}K_t\text{-models})$

$$\begin{aligned} &\leq t^n \exp\left\{-\frac{(t - 1)^2}{e} (1 - p)^{(n-1)/(t-1)}\right\} \\ &= \exp\left\{(n \log t) - \frac{(t - 1)^2}{e} (1 - p)^{(n-1)/(t-1)}\right\}. \end{aligned}$$

Recall that $t - 1 = (1 + \varepsilon) \frac{n-1}{\log_b n}$. Thus, $(1 - p)^{\frac{n-1}{t-1}} = n^{-1/(1+\varepsilon)}$ and

$$\begin{aligned} n \log t - \frac{(t - 1)^2}{e} (1 - p)^{(n-1)/(t-1)} \\ \leq n \log n - \frac{(1 + \varepsilon)^2}{e(\log_b n)^2} (n - 1)^2 n^{-1/(1+\varepsilon)}. \end{aligned}$$

Since $2 - \frac{1}{1+\varepsilon} > 1 + \frac{\varepsilon}{2}$ (assuming $\varepsilon < 1$), the right-hand side tends to $-\infty$ as $n \rightarrow \infty$. Thus, the expected number of dominating pseudo- K_t -models, and so the probability that there is a dominating pseudo- K_t -model is $o(1)$, as required. $\square$

5 | Structure

This section explores the structure of graphs with no dominating K_t -model, and concludes a lower bound on the independence number of such graphs. The proof follows that of Duchet and Meyniel [5] for K_t -minor-free graphs.

Lemma 12 ([5]). *Every non-empty connected graph has a connected dominating set D and an independent set $I \subseteq D$ with $|D| = 2|I| - 1$.*

Proof. Let D be the largest connected set in G such that D contains an independent set I with $|D| = 2|I| - 1$. This is well-defined since $D = I = \{v\}$ satisfies these properties for any vertex v . Suppose for the sake of contradiction that D is not dominating. Since G is connected, there exists an edge xy in G with $\text{dist}_G(x, D) = 1$ and $\text{dist}_G(y, D) = 2$. Thus, $D' := D \cup \{x, y\}$ is a connected set and $I' := I \cup \{y\}$ is an independent set in D' with $|D'| = 2|I'| - 1$, contradicting the choice of D and I . Hence D is dominating. $\square$

A *partition* of a graph G is a partition of $V(G)$ into non-empty sets. Each element of a partition is called a *part*. Let $\mathcal{P}$ be a partition of a graph G . The *quotient* of $\mathcal{P}$ is the graph with vertex-set $\mathcal{P}$ where distinct $A, B \in \mathcal{P}$ are adjacent if and only if there is an edge of G between A and B . If H is isomorphic to the quotient of $\mathcal{P}$, then $\mathcal{P}$ is called an H -*partition* of G . We say $\mathcal{P}$ is *connected* if each part induces a connected subgraph of G , in which case the quotient is a minor of G .

The *depth* of a vertex v in a tree rooted at r is the number of vertices in the vr -path in T . The *closure* of a rooted tree T is the graph $\hat{T}$ with vertex-set $V(T)$, where vw is an edge of $\hat{T}$ if and only if v is an ancestor or descendent of w in T . A graph that is isomorphic to the closure of a tree is called *trivially perfect*, a *comparability graph of a tree*, an *arborescent comparability graph*, or a *quasi-threshold graph* [11]. They are the graphs that contain neither P_4 nor C_4 as induced subgraphs.

Theorem 13. *For every connected graph G there is a tree T rooted at r such that G has a connected $\hat{T}$ -partition $\mathcal{P}$ where:*

- for every part $P \in \mathcal{P}$ there is an independent set I in $G[P]$ with $|P| = 2|I| - 1$ and
- for every leaf v of T , the sequence of parts in $\mathcal{P}$ from r to v define a dominating complete graph model in G .

Proof. We proceed by induction on $|V(G)|$. The case $|V(G)| = 1$ is trivial. By Lemma 12, G has a connected dominating set D and an independent set $I \subseteq D$ with $|D| = 2|I| - 1$. Let $G_1, \dots, G_c$ be the components of $G - D$. By induction, for each $i \in \{1, \dots, c\}$, there is a tree T_i rooted at r_i such that G_i has a $\hat{T}_i$ -partition $\mathcal{P}_i$ such that:

- for every part $P \in \mathcal{P}_i$ there is an independent set I in $G_i[P]$ with $|P| = 2|I| - 1$ and
- for every leaf v of T_i , the sequence of parts in $\mathcal{P}_i$ from r_i to v defines a dominating complete graph model in G_i .

Let T be the tree obtained from the disjoint union of $T_1, \dots, T_c$ by adding a new root vertex r adjacent to $r_1, \dots, r_c$. So $\hat{T}$ is obtained

from the disjoint union of $\hat{T}_1, \dots, \hat{T}_c$ by adding vertex r , adjacent to every other vertex. Let $\mathcal{P}$ be the $\hat{T}$ -partition of G obtained by associating D with r . Since D is dominating, for every leaf v of T , the sequence of parts in $\mathcal{P}$ from r to v defines a dominating complete graph model in G . $\square$

Theorem 14. *For $t \geq 2$ and every graph G with no dominating K_t -model, there is an integer $h \leq t - 1$ and an h -colourable induced subgraph of G on at least $\frac{|V(G)| + h}{2}$ vertices. In particular,*

$$\alpha(G) \geq \frac{|V(G)| + t - 1}{2t - 2}.$$

Proof. We may assume G is connected. Apply Theorem 13 to G . Let T be the resulting tree, and let $\mathcal{P}$ be the resulting $\hat{T}$ -partition of G . Let h be the maximum depth of a node in T . Since G has no dominating K_t -model, $h \leq t - 1$. Each part $A \in \mathcal{P}$ contains an independent set I_A of size $(|A| + 1)/2$. Let X be the union of the I_A . Then

$$|X| \geq \sum_{A \in \mathcal{P}} \frac{|A| + 1}{2} = \frac{|V(G)| + |\mathcal{P}|}{2} \geq \frac{|V(G)| + h}{2}.$$

Colour each vertex in I_A by the depth of the corresponding node in T . We obtain a proper h -colouring of X . Taking the largest colour class in X ,

$$\begin{aligned} \alpha(G) &\geq \frac{|V(G)| + h}{2h} = \frac{|V(G)|}{2h} + \frac{1}{2} \geq \frac{|V(G)|}{2t - 2} + \frac{1}{2} \\ &= \frac{|V(G)| + t - 1}{2t - 2}. \end{aligned} \quad \square$$

6 | Average Degree

This section considers the maximum average degree of a graph with no dominating K_t -model. Theorem 11 implies the following.

Corollary 15. *For any $\varepsilon > 0$ and for sufficiently large t , there is a graph G with no dominating K_t -model and average degree at least $(1 - \varepsilon)t \ln t$.*

Proof. Note that $\ln\left(\frac{1}{1-x}\right)/x \rightarrow 1$ as $x \rightarrow 0$. Fix $\varepsilon > 0$ small and let $p > 0$ be small enough so that $\ln\left(\frac{1}{1-p}\right) < \left(1 + \frac{\varepsilon}{3}\right)p$. For large n , with high probability, $G(n, p)$ has average degree at least $(1 - \varepsilon^2)pn$, and by Theorem 11,

$$\begin{aligned} \text{domhad}(G(n, p)) &\leq \left(1 + \frac{\varepsilon}{3}\right) \frac{n}{\log_b n} = \left(1 + \frac{\varepsilon}{3}\right) \frac{n \ln b}{\ln n} \\ &< \left(1 + \frac{\varepsilon}{3}\right)^2 \frac{pn}{\ln n} < (1 + \varepsilon) \frac{pn}{\ln n}. \end{aligned}$$

Pick an instance G of $G(n, p)$ satisfying both these conditions. Then, G has no dominating K_t -model where $t := (1 + \varepsilon) \frac{pn}{\ln n}$. Finally,

$$(1 - \varepsilon)t \ln t \leq (1 - \varepsilon)(1 + \varepsilon) \frac{pn}{\ln n} \cdot \ln n = (1 - \varepsilon^2)pn,$$

which is at most the average degree of G , as required. $\square$

Mader [12] first showed that every graph with sufficiently high average degree contains a K_t -minor. The proof method actually shows the following:

Theorem 16. *For $t \geq 2$ every graph with average degree at least 2^{t-2} has a dominating K_t -model.*

Proof. We proceed by induction on $n + t$ with the following hypothesis: for every connected n -vertex graph G with average degree at least 2^{t-2} and for every vertex v of G , there exists a dominating K_t -model $(T_1, \dots, T_t)$ in G with $v \in V(T_1)$. This implies the claim since every graph with average degree at least 2^{t-2} has a connected component with average degree at least 2^{t-2} .

The $t = 2$ case holds with $T_1 = G[\{v\}]$ and $T_2 = G[\{w\}]$ where w is any neighbour of v (which exists since G is connected with average degree at least 1). If $n \leq 2^{t-2} + 1$ then G is complete, and the result holds trivially. Now, assume that $t \geq 3$ and $n > 2^{t-2} + 1$. Let G be a connected n -vertex m -edge graph G with average degree at least 2^{t-2} , and let v be a vertex of G . So $2m \geq 2^{t-2}n$.

For each neighbour w of v , let d_w be the number of common neighbours of v and w . Suppose that $d_w \leq 2^{t-3} - 1$ for some neighbour w of v . Let G' be the graph obtained from G by contracting vw into a new vertex v' . So G' is connected, $|E(G')| = m - 1 - d_w$ and $|V(G')| = n - 1$. Thus, G' has average degree

$$\begin{aligned} \frac{2(m - 1 - d_w)}{n - 1} &\geq \frac{2^{t-2}n - 2(1 + d_w)}{n - 1} \geq \frac{2^{t-2}n - 2^{t-2}}{n - 1} \\ &= 2^{t-2}. \end{aligned}$$

By induction, there exists a dominating K_t -model $(T_1, \dots, T_t)$ in G' with $v' \in V(T_1)$. Let T'_1 be obtained from T_1 by replacing v' by vw , and replacing each edge $v'x$ in T_1 by vx or wx (one of which must exist). By construction, every vertex adjacent to v' in G' is adjacent to v or w in G . Thus $(T'_1, T_2, \dots, T_t)$ is a dominating K_t -model in G with $v \in V(T'_1)$, as desired.

Now, assume that every edge incident to v is in at least 2^{t-3} triangles. Since G is connected and with at least two vertices, there is at least one edge incident to v . Thus, $G[N_G(v)]$ has minimum degree at least 2^{t-3} . By induction, $G[N_G(v)]$ has a dominating K_{t-1} -model $(T_1, \dots, T_{t-1})$. Thus, $(\{v\}, T_1, \dots, T_{t-1})$ is the desired dominating K_t -model in G . $\square$

We conjecture that average degree $\Omega(t \log t)$ is enough to force a dominating K_t -model (and Corollary 15 shows this would be best possible). As a step towards this conjecture, we show it is true for graphs with linear minimum degree. The following proofs make no attempt to optimise constants. The first lemma is well-known.

Lemma 17. *Every connected graph G with n vertices and minimum degree $\delta > \ln(2n)$ has a connected dominating set on at most $\frac{6n \ln(2n)}{\delta}$ vertices.*

Proof. Let $p := \ln(2n)/\delta < 1$. For each vertex v , choose v independently at random with probability p . Let A be the set of chosen vertices. Let B be the set of vertices v such that $N_G(v) \cap A = \emptyset$. Thus, $\mathbb{P}(v \in B) = (1 - p)^{\deg(v)} < e^{-p\delta} = \frac{1}{2n}$. Thus, $\mathbb{E}(|B|) < \frac{1}{2}$, and by Markov's Inequality, $\mathbb{P}(|B| \geq 1) \leq \frac{\mathbb{E}(|B|)}{1} < \frac{1}{2}$. Now consider $|A|$: $\mathbb{E}(|A|) = pn$ and so, by Markov's Inequality, $\mathbb{P}(|A| \geq 2pn) \leq \frac{1}{2}$. A union bound gives $\mathbb{P}(|B| \geq 1 \text{ or } |A| \geq 2pn) < 1$. Hence, there exists $A \subseteq V(G)$ with $|A| < 2pn = \frac{2n \ln(2n)}{\delta}$ and $B = \emptyset$. Since $B = \emptyset$, A is a dominating set. The result follows, since Duchet and Meyniel [5] showed that if a connected graph has a dominating set on k vertices, then it has a connected dominating set on at most $3k - 2$ vertices. $\square$

The next lemma is similar to several results in the literature on K_t -minors [13, 14].

Lemma 18. *Fix $c \in (0, 1)$ and n such that $cn > \ln(2n)$. Then, every graph G with at most n vertices and minimum degree at least $cn + 6c^{-1}t \ln(2n)$ has a dominating K_t -model.*

Proof. We proceed by induction on t . The case $t = 1$ is trivial. We may assume that G is connected, since any component G' of G has $\delta(G') \geq \delta(G) \geq cn + 6c^{-1}t \ln(2n)$ and $|V(G')| \leq |V(G)| \leq n$. Since G has minimum degree at least cn , by Lemma 17, G has a connected dominating set A with $|A| \leq 6c^{-1} \ln(2n)$. Let $G' := G - A$. So G' has at most n vertices and minimum degree at least $cn + 6c^{-1}(t - 1) \ln(2n)$. By induction, G' has a dominating K_{t-1} -model $(A_1, \dots, A_{t-1})$. Since A is dominating, $(A, A_1, \dots, A_{t-1})$ is a dominating K_t -model in G . $\square$

Proposition 19. *For any $c \in (0, 1)$ and sufficiently large integers n, t with $n \geq c^{-2}t \log_2 t$, every graph with n vertices and minimum degree at least $2cn$ has a dominating K_t -model.*

Proof. We may assume that $t \geq (2c^{-2} \log_2 t)^6$. Thus

$$\log_2 t \geq \ln t \geq 6 \ln(2c^{-2} \log_2 t).$$

Since $\frac{n}{\ln(2n)}$ is increasing and $n \geq c^{-2}t \log_2 t$,

$$\frac{c^2 n}{\ln(2n)} \geq \frac{t \log_2 t}{\ln(2c^{-2}t \log_2 t)} \geq 6t.$$

Therefore

$$2cn \geq cn + 6c^{-1}t \ln(2n).$$

The result follows from Lemma 18 with minimum degree $2cn$, since we may assume that $cn > \ln(2n)$. $\square$

Here is further evidence for the conjecture.

Proposition 20. For sufficiently large t and $d = 4t \ln t$, every d -regular graph G contains a dominating pseudo- K_t -model.

Proof. Uniformly and randomly put each vertex of G into one of t parts. For each vertex v , let B_v be the event that v has no neighbour in some part. Note that $\mathbb{P}(B_v)$ equals the probability that a coupon collector has not succeeded by time $4t \ln t$, which is at most t^{-3} by standard tail estimate for the coupon collector problem [15]. Note that B_v depends only on the parts where the neighbours of v were placed, and so B_v is independent of $(B_u : \text{dist}(u, v) \geq 3)$. There are at most d^2 vertices within distance 2 of v (other than v). Since $4t^{-3}d^2 < 1$ for sufficiently large t , the Lovász Local Lemma [16] implies that no B_v occurs with positive probability. That is, there exists a partition of $V(G)$ such that every vertex in G has a neighbour in every part. This partition defines a dominating pseudo- K_t -model. $\square$

This proof still works if all degrees are close to each other, but does not work if the degrees vary wildly.

7 | Chromatic Number

This section considers the chromatic number of graphs with no dominating K_t -model. Mader [12] showed that every K_t -minor-free graph is 2^{t-2} -colourable. The proof generalises as follows:

Theorem 21. For $t \geq 2$, every graph with no dominating K_t -model is 2^{t-2} -colourable.

Proof. We proceed by induction on t . Every graph G with no dominating K_2 -model has no edges, implying G is 1-colourable. Now assume that $t \geq 3$, and the result holds for $t - 1$. Let G be a graph with no dominating K_t -model. We may assume that G is connected. Let r be any vertex in G . For $i \geq 0$, let $V_i := \{v \in V(G) : \text{dist}_G(v, r) = i\}$. So $V_0 = \{r\}$, and for $i \geq 1$, every vertex in V_i has a neighbour in V_{i-1} . If for some $i \geq 1$, there is a dominating K_{t-1} -model $(T_1, \dots, T_{t-1})$ in $G[V_i]$, then $(G[V_0 \cup \dots \cup V_{i-1}], T_1, \dots, T_{t-1})$ is a dominating K_t -model in G . Thus for every $i \geq 0$, there is no dominating K_{t-1} -model in $G[V_i]$. By induction, each $G[V_i]$ is 2^{t-3} -colourable. Use the same set of 2^{t-3} colours for $\bigcup(V_i : i \text{ even})$ and use a disjoint set of 2^{t-3} colours for $\bigcup(V_i : i \text{ odd})$. Since there is no edge between V_i and V_j with $j \geq i + 2$, we obtain a 2^{t-2} -colouring of G . $\square$

Note that Theorem 16 implies that every graph with no dominating K_t -model is $(2^{t-2} - 1)$ -degenerate and thus 2^{t-2} -colourable, which provides an alternative proof of Theorem 21.

The proof of Theorem 21 in conjunction with Corollary 9 in the base case shows:

Theorem 22. For $t \geq 4$, every graph with no dominating K_t -model is $3 \cdot 2^{t-4}$ -colourable.

Now consider the chromatic number of random graphs. For constant $p \in (0, 1)$, the asymptotic value of $\chi(G(n, p))$

was determined independently by Bollobás [17] and Matula and Kučera [18]. They proved that, with high probability,

$$\chi(G(n, p)) = \left(\frac{1}{2} + o(1)\right) \frac{n}{\log_b n}, \quad (2)$$

where $b := \frac{1}{1-p}$. See Heckel [19] for the most precise estimates currently known.

Bollobás et al. [6] showed that for fixed $p \in (0, 1)$, with high probability,

$$\text{had}(G(n, p)) = (1 + o(1)) \frac{n}{\sqrt{\log_b n}}.$$

It follows that Hadwiger's conjecture comfortably holds for almost every graph.

Theorem 11 and (2) imply that:

Corollary 23. Almost every graph G satisfies $\chi(G) \leq \text{domhad}(G)$.

It is curious that the chromatic number and dominating Hadwiger number asymptotically differ only by a factor of 2, compared to a factor of $\sqrt{\log_2 n}$ in the case of the Hadwiger number.

We finish this section with the following amusing observation. Let G be a graph with $\chi(G) = t$. Choose a proper colouring of G with colours $1, \dots, t$ minimising $\sum_{v \in V(G)} \text{col}(v)$. Then for $1 \leq i < j \leq t$, every vertex of colour j is adjacent to a vertex of colour i . In particular, the colour classes form a dominating pseudo- K_t -model. This says that the 'dominating pseudo-Hadwiger conjecture' is true!

8 | Open Problems

We finish with a number of open problems.

- Several authors [20–23] have noted that graphs G with independence number $\alpha(G) = 2$ are a key unsolved case of Hadwiger's conjecture. Plummer et al. [20] showed that Hadwiger's conjecture holds for this class if and only if every n -vertex graph G with $\alpha(G) = 2$ has a K_t -minor with $t \geq \left\lfloor \frac{n}{2} \right\rfloor$. The following natural question arises: Does every n -vertex graph with $\alpha(G) = 2$ have a dominating K_t -model with $t \geq \left\lfloor \frac{n}{2} \right\rfloor$? Note that if $\alpha(G) = 2$ and G has no dominating K_t -model, then $2 \geq \alpha(G) \geq \frac{n+t-1}{2t-2}$ by Theorem 14, implying $n \leq 3t - 3$. That is, if $\alpha(G) = 2$ then G has a dominating K_t -model, where $t \geq \left\lfloor \frac{n}{3} \right\rfloor$.
- The following potential strengthening of the four-colour theorem is open: Is every graph with no dominating K_5 -model 4-colourable?

- Several authors have studied the choosability of K_t -minor-free graphs; see [4, 24] for upper bounds, and see [25, 26] for lower bounds. What is the maximum choosability of the class of graphs with no dominating K_t -model? Theorem 16 implies an upper bound of 2^{t-2} . Theorem 7 implies that every graph with no dominating K_4 -model is 3-choosable. The case $t = 5$ is of particular interest. Thomassen [27] showed that every planar graph is 5-choosable. Generalising this result, Škrekovski [28] showed that every K_5 -minor-free graph is 5-choosable (see [29, 30] for alternative proofs). Is every graph with no dominating K_5 -model 5-choosable?
- Does every graph with no dominating K_t -model have fractional chromatic number at most $2t - 2$? Reed and Seymour [31] proved this for K_t -minor-free graphs.
- Is there a rough structure theorem for graphs with no dominating K_t -model (in the spirit of Robertson and Seymour's rough structure theorem for K_t -minor-free graphs [32] and Grohe and Marx's rough structure theorem for K_t -topological-minor-free graphs [33])?

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Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

Endnotes

¹For a graph G , a set $A \subseteq V(G)$ is *dominating* in G if every vertex in $V(G) \setminus A$ has a neighbour in A . A set $A \subseteq V(G)$ is *connected* if $G[A]$ is connected.

²The *Erdős-Rényi random graph* $G(n, p)$ is the n -vertex graph in which each edge is present with probability p independently of all other edges.

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